Learning Chaotic Dynamics of the Lorenz System with Neural ODEs and Physics-Informed Neural Networks

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***Abstract*—The Lorenz system is a hallmark of chaotic dynamics, characterized by extreme sensitivity to initial conditions. Traditional numerical solvers struggle with long-term trajectory prediction due to error accumulation. This project investigates the efficacy of two advanced Scientific Machine Learning (SciML) approaches: Neural Ordinary Differential Equations (Neural ODEs) and Physics-Informed Neural Networks (PINNs). Their performance is evaluated against baseline Multi-Layer Perceptrons (MLPs) and the classic 4th-order Runge-Kutta (RK4) method. The results demonstrate how encoding physical priors or continuous-time dynamics improves prediction stability, while also highlighting the inherent limitations of ML models in overcoming long-term chaotic divergence.**

I. Introduction

Chaotic systems, such as the Lorenz attractor, present significant challenges for predictive modeling due to their inherent sensitivity to initial conditions—commonly known as the butterfly effect. Traditional numerical methods, such as the Euler and Runge-Kutta (RK) families, often suffer from rapid error accumulation when predicting long-term trajectories.

Recently, Scientific Machine Learning (SciML) has emerged as a promising paradigm. While standard deep learning architectures (e.g., MLPs or standard RNNs) often fail to respect underlying physical laws, methods like Neural Ordinary Differential Equations (Neural ODEs) and Physics-Informed Neural Networks (PINNs) offer robust alternatives [1], [2]. In this report, we evaluate the capability of Neural ODEs and PINNs to learn and predict the chaotic dynamics of the Lorenz system, comparing their trajectory error and long-term stability against traditional baseline methods.

II. Methodology

A. The Lorenz System and Data Generation

The Lorenz system is defined by the following set of coupled nonlinear ordinary differential equations:

$$\begin{aligned} \frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z \end{aligned} \tag{1}$$

For our experiments, we use standard chaotic parameters ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$). Ground truth data is generated using the RK4 method to serve as both training data and the baseline for evaluation.

B. Neural ODE Architecture

Neural ODEs parameterize the derivative of the hidden state using a neural network, treating the forward pass as the solution to an initial value problem (IVP) [1]. We employ a neural network f_θ to approximate the continuous-time dynamics of the Lorenz system.

C. Physics-Informed Neural Network (PINN)

For the PINN implementation, the loss function is formulated as a combination of data-driven loss and physics-informed loss based on Eq. 1 [2]. The total loss $\mathcal{L}$ is defined as:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics} \tag{2}$$

where $\mathcal{L}_{physics}$ penalizes the residual of the governing differential equations, computed efficiently via automatic differentiation.

III. Results

To rigorously evaluate the models, we simulated the Lorenz system ($\sigma = 10, \rho = 28, \beta = 8/3$) over a time domain of $t \in [0, 20]$ with a step size of $\Delta t = 0.01$, generating 2000 discrete data points via the RK45 method to serve as the ground truth. The models were trained on a subset of this data and tested on their ability to autonomously forecast the trajectory given an initial condition.

A. Baseline MLP: Catastrophic Divergence

The baseline Multi-Layer Perceptron was trained to learn the discrete step-to-step mapping $x_{t+1} = f(x_t)$. As illustrated in Fig. 1, while the MLP achieved a low Mean Squared Error (MSE) during single-step training, it failed entirely during autoregressive rollout. The predicted trajectory rapidly diverged from the true chaotic attractor and collapsed into a non-physical micro-orbit. This confirms that naive feedforward architectures lack the temporal continuous inductive bias required for dynamical systems.

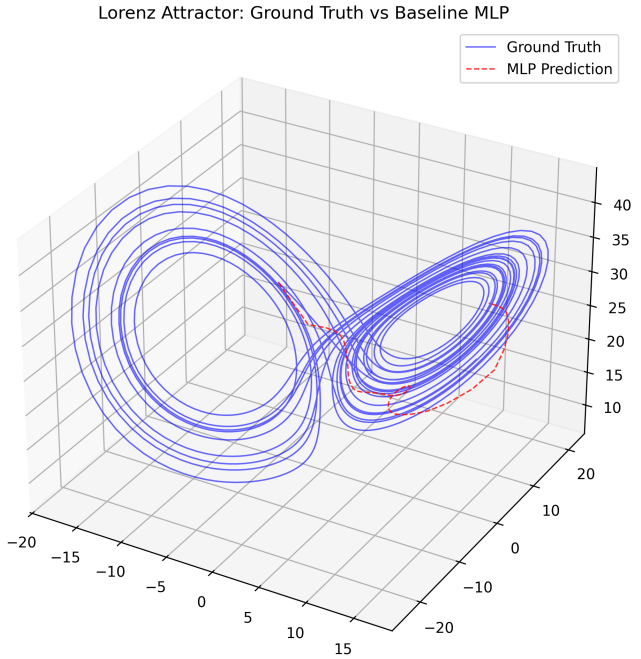

Fig. 1. Autoregressive forecasting of the baseline MLP. The model completely fails to capture the chaotic dynamics, resulting in catastrophic divergence.

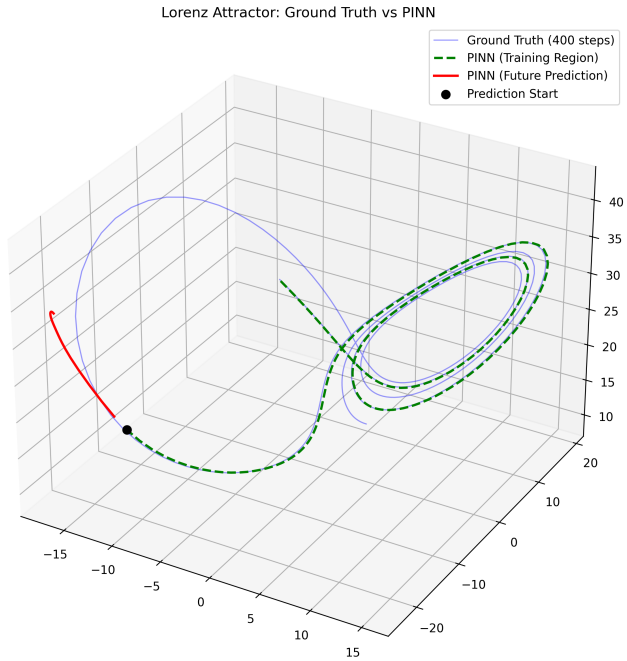

Fig. 2. Autoregressive extrapolation of the PINN. The model perfectly captures the physical dynamics within the training region (green dashed line) but exponentially diverges during future prediction (red solid line) due to the inherent sensitivity of chaotic systems.

B. PINN: Overfitting and Extrapolation Failure

We incorporated the governing differential equations of the Lorenz system into the loss function to train a continuous-time Physics-Informed Neural Network (PINN). As shown in Fig. 2, the PINN successfully learned the dynamics within the training window (green dashed line), perfectly interpolating the ground truth. However, beyond the prediction start point, the extrapolation phase (red solid line) rapidly diverged from the true chaotic attractor. This demonstrates that while PINNs can interpolate accurately within observed data regimes, relying solely on physics residuals without continuous data assimilation is insufficient for the long-term forecasting of highly sensitive chaotic systems.

C. Neural ODE: Limit Cycles

The Neural Ordinary Differential Equation framework demonstrated the most robust macroscopic behavior (Fig. 3). By parameterizing the derivative and utilizing an adjoint ODE solver, the model successfully constrained its predictions within the physical manifold. However, constrained by the MSE loss function and the exponential divergence of chaos, the model eventually collapsed into a stable limit cycle rather than replicating the aperiodic lobe-switching behavior.

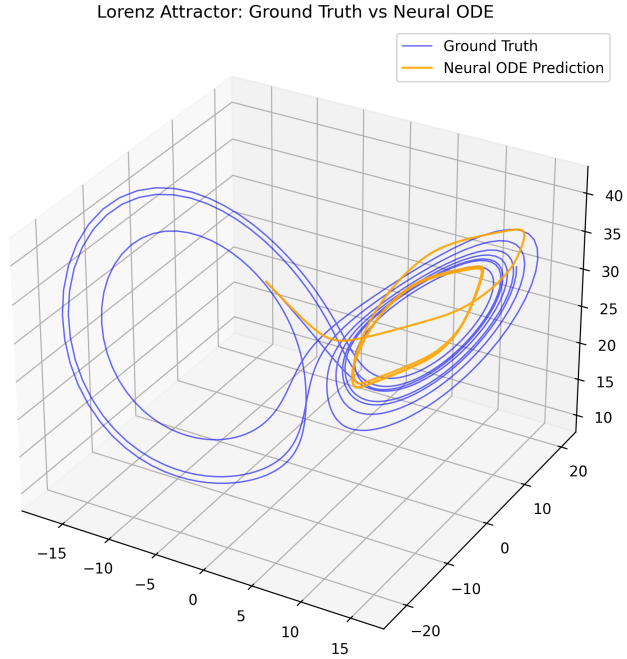

Fig. 3. Long-term prediction using Neural ODE. The model restricts the trajectory to the correct manifold but eventually falls into a limit cycle due to MSE penalties.

IV. Discussion

Our comparative analysis yields a critical insight into Scientific Machine Learning: while embedding continuous-time dynamics (Neural ODEs) or physical priors (PINNs) improves short-term interpolation compared to naive deep learning,

long-term extrapolation in chaotic systems like the Lorenz attractor remains a formidable challenge.

The primary limitation stems from the objective function. Standard MSE loss severely penalizes the model when a small temporal phase shift occurs during lobe transitions. To minimize this catastrophic penalty, the network learns a ”safe” conservative dynamic, often manifesting as a limit cycle. This reveals a fundamental misalignment between pointwise loss metrics and the topological objectives of chaotic forecasting.

Future work must transition from trajectory-matching to invariant-measure-matching. Exploring chaos-specific loss functions, integrating reservoir computing with Neural ODEs, or employing curriculum learning strategies (e.g., time-marching) are essential next steps to fully capture the rich, aperiodic nature of chaotic dynamics.

REFERENCES

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